

Using Large Language Models to Contextualize Smiling in Psychotherapy Research

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Abstract: The meaning of a smile during conversation depends on what is being said, yet psychotherapy research has historically analyzed facial movements independently of context. We introduce a methodology that links both partners' smiling in a dyadic setting to the sentiment of a speech turn (positive, neutral, negative). Using open-weight models on local hardware only, we transcribed and diarized attachment interviews, annotated the sentiment of the participant's speech turns with a large language model (LLM), and analyzed individual

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smiling and smile synchrony as a function of that sentiment. We applied linear mixed models to test for differences in smiling according to sentiment. Across 70 interviews and 1,294 speech turns, only the interviewer's smiling differed significantly according to the sentiment of what the participant was saying (interviewer $d = .51$, $q < .001$; participant $d = .05$, $q = .165$); the pair smiled simultaneously more often during positive than negative turns ($d = .21$, $p < .001$). By conditioning smiling on what is being said — sentiment annotated from the conversation as it unfolds rather than created by experimental instruction — our method enables the kind of context-sensitive analysis contemporary theories call for. The pipeline is open and reproducible, runs entirely on local hardware so that no data leave the site, and can be easily modified for any annotation scheme or clinical setting.

Keywords: Nonverbal behavior, Large Language Models, psychotherapy, interpersonal synchrony, Duchenne smile

Introduction

Research on nonverbal behavior during psychotherapy has a long tradition in psychotherapy research (Bänninger-Huber & Huber, 2017; Luo & Levendosky, 2025; Sayda et al., 2026; Watson, 2025; Zilcha-Mano et al., 2026), yet a central methodological challenge has persisted throughout: nonverbal behavior, typically operationalized as facial movements, has consistently been measured independently of the spoken content. This separation is critical, because the meaning of a given facial configuration varies with the communicative context and sentiment in which it occurs (Barrett, 2022; Barrett et al., 2019). Smiling is the preeminent case. It is among the most frequent and relationally salient facial behaviors in the therapeutic encounter (Girard et al., 2014), and even its most well-specified form, the Duchenne smile (Ekman et al., 1990), does not cleanly map to a single affective state (Girard et al., 2021). A client's smile may convey genuine warmth as a painful memory gives way to relief, yet the same configuration can signal unease, appeasement, or embarrassment when the surrounding talk is distressing. Without access to what is being said, no measurement procedure can distinguish between these instances.

The dominant methodological paradigm has been unable to capture this context sensitivity, because the nonverbal signal is aggregated across the entire session. The behavior itself is coded automatically, whether by Motion Energy Analysis (MEA) (Ramseyer, 2020) for body movement or by OpenFace (Baltrušaitis et al., 2016) for facial action units (AUs), both of which yield frame-by-frame time series. Synchrony is then computed, for instance by averaging windowed cross-lagged correlations across the session (Kleinbub & Ramseyer, 2021). The context is discarded at the final step, where these estimates are pooled over a whole session or a fixed opening segment and the resulting single value is related to a post-session score, often alliance or outcome (Ramseyer & Tschacher, 2011; Tschacher et al., 2014). Aggregating synchrony in this way means treating every moment of a session as interchangeable, so that a synchronous movement while a client recounts a loss and one during light rapport-building contribute equally to the same number. Yet clinical work on smiling and laughter in dyadic interaction shows how strongly their function depends on the moment in which they occur: whether they regulate negative affect or express positive affect, whether the partner reciprocates

them, and where in the session they fall (Bänninger-Huber & Huber, 2017; Bänninger-Huber & Salvenauer, 2023). When synchrony is aggregated across the session, the semantic context that gives a nonverbal signal its meaning is averaged out of the measure, and precisely the effects that matter most risk canceling out.

Large Language Models for sentiment annotation and its clinical value

We propose to draw on the insight that the context needed to resolve this problem can be extracted from the spoken content of the interaction. What has been missing is a way to quantify it at scale, as manual coding of situational meaning is too labor-intensive for turn-by-turn analysis. Large language models (LLMs) close this gap. We present a method in which an instruction-tuned, open-weight LLM annotates the situational valence of each speech turn, yielding a semantic annotation against which individual and synchronous smiling can be quantified. Little prior work exists in this domain: applications of LLMs in psychotherapy research have been oriented almost exclusively toward the verbal modality (Eberhardt, Schaffrath, et al., 2025; Stade et al., 2024), and where LLMs have been paired with nonverbal signals, the work has come from human–computer interaction and aimed at building conversational agents (Galland et al., 2025; Schmidmaier et al., 2025; Wessler et al., 2024) as opposed to analyzing the nonverbal behavior of interacting people.

Enabling context-sensitive analysis of smiling holds substantial implications for both clinical and theoretical psychotherapy research. Robust evidence exists on the relationship between therapeutic alliance and treatment outcome (Cuijpers et al., 2019; Del Re et al., 2012; Flückiger et al., 2018; McLeod, 2011; Wampold, 2015), with interpersonal synchrony posited as a possible mediating mechanism (Koole et al., 2020). Yet empirical findings on synchrony and alliance remain mixed (Atzil-Slonim et al., 2023; Gregorini et al., 2025; Lucherini Angeletti et al., 2025). Some studies report that more synchrony goes with a stronger alliance, others that it goes with a weaker one, and pooling the two directions leads to no combined effect (Jennissen et al., 2025); expected if synchrony were beneficial at some moments and not at others.

Recent theoretical work on flexible multimodal synchrony offers a viable explanation for these inconsistencies: Gordon et al. (2025) argue that synchrony is not a uniform meta-stable

state of a dyad but a dynamical process in which the conversation creates opposing demands on affiliation (through synchronization) and autonomy (through de-synchronization). What predicts relational outcomes is not the amount of interpersonal synchrony but the degree of relational *flexibility*, i.e., the capacity to change synchrony as the context changes. Initial evidence supports this account: examining five synchrony modalities in patient-therapist dyads, Sayda et al. (2026) found that the flexibility of facial-expression synchrony moderated its association with session outcome, with higher synchrony predicting better outcomes only when synchrony was flexible. In that study, however, flexibility was defined simply as how much synchrony fluctuated over the course of a session, without linking it to what was being discussed at any given moment.

Previous studies examining how context and situational valence influence synchrony are both sparse and methodologically heterogeneous. Kimura and Daibo (2006) found that observers perceived synchrony in conversations about both positive and negative emotional episodes, though more strongly for positive ones. Altmann (2011) showed that children’s body-movement synchrony was higher in neutral than in conflict situations, and Tschacher et al. (2014) found in a larger dyadic experiment that body-movement synchrony positively predicted positive affect and reduced negative affect. Closest to the present focus, Kykyri et al. (2025) reported that in a single couple-therapy session, movement synchrony followed emotional valence. These studies, however, treat sentiment as a whole-episode property (for instance, set by design, self-report, or global coding), leaving little room to account for the moment-to-moment flexibility articulated by Gordon et al. (2025). Quantifying facial synchrony turn by turn, against the valence of each turn’s content, directly addresses these limitations.

In the presented methodology, an LLM annotates each participant’s speech turn as positive, neutral, or negative, on the basis of the verbal content alone. Here we apply it to attachment interviews (Andreas et al., 2022), analyzing the frequency and intensity of Duchenne smiling in both the participant and the interviewer, and the synchrony between them, as a function of this sentiment. Our objective is twofold. First, we introduce the methodology: an open-weight pipeline that runs entirely on local hardware, is highly customizable to other research designs and fully open-source. Second, we illustrate it, characterizing how participant and interviewer Duchenne smiling differ across sentiment categories (Analysis 1), and whether

participant–interviewer smile synchrony is itself sentiment-specific, estimated both as the frequency of simultaneous smiling (Analysis 2a) and as the cross-lagged correlation of smile intensity (Analysis 2b).

Method

Participant recruitment procedure

Our initial sample comprises 83 dyads that originate from an observational study conducted at the Medical School Berlin (MSB) by Altmann and Volkert (2023) (also appears in Rasche et al. (n.d.), in preparation). The recruitment strategy utilized digital and physical flyers across the university, university email distribution lists, social media networks, and outreach through supporting psychotherapeutic clinics. Participants had to meet the following inclusion criteria: 1) Be 18 years of age or older and not under any legal guardianship. 2) Possess adequate sensory capacity to accurately perceive the spoken and nonverbal behavior of the conversational partner. 3) Have sufficient German language skills. 4) Willingness to complete the questionnaires and participate in an approximately 60-minute interview. 5) Provide informed consent for the pseudonymized audio and video recordings to be analyzed for scientific purposes. 6) Be fully informed regarding the study's objectives, procedures, and duration. Following eligibility screening, participants provided their contact details and received a personalized link to complete the baseline assessments (T0) via the secure, ISO27001-certified online platform SoSci Survey. Once the baseline questionnaires were completed, an interview appointment was scheduled. To minimize variance related to gender constellation of dyads, all interviews were conducted exclusively by female interviewers. The interviewers were advanced undergraduate or graduate students in psychology who received comprehensive theoretical and practical training from two licensed psychologists before data collection. We note that we had to reduce our initial sample of 83 to 70 dyads due to requirements set by our language model. We elaborate on these sample reductions in [Sample characteristics](#).

Study design

The video study consisted of three parts: a personality interview, an attachment interview, and a solution-focused intervention (see [Table S1](#)). In the present study, we use only the recording from the attachment interview (Interview 2). We intentionally restrict our sample to this particular interview, as our study design requires an LLM to assign a sentiment category (negative, neutral, positive, or other) to each participant’s speech turn (see details in [Language processing](#)). During the attachment interview, the participant disclosed positive and negative attachment-related experiences, so that the sentiment of a speech turn could be assigned from the relationship being described. This condition was not met in the personality functioning interview or during the solution-focused intervention, and the majority of turns would therefore fall into the *other* category. The attachment interview was thus the only interview in which the conversation was of a nature that permitted this category assignment.

Instruments

Brief Reflective Functioning interview (BRFI). The attachment interview used in the present study was the semi-structured BRFI, which invites the participant to reflect on their relationship with a self-chosen parent (Rudden et al., 2006). The BRFI demonstrates excellent internal consistency ($\alpha = .92$), high interrater reliability ($ICC = .85$), and strong convergent validity with the Adult Attachment Interview (Andreas et al., 2022).

Data preparation and feature extraction

Our data preparation pipeline operates across two complementary modalities, as illustrated in [Figure 1](#). The textual modality transforms raw audio recordings into sentiment-annotated speech turns through a sequential chain of diarization, transcription, translation, and LLM-based sentiment classification. In parallel, the visual modality extracts frame-level facial AU intensities and aligns them to the speech-turn structure. Each annotated speech turn carries the start and end time of the participant’s speech, and its sentiment label is shared across every video frame within that interval. The text workflow therefore yields a frame-level coding of verbal sentiment, aligned in time with the frame-level AU estimates produced by the facial workflow.

All subsequent measures are computed over the frames that a single sentiment label spans. The following sections describe the core components of our pipeline in detail.

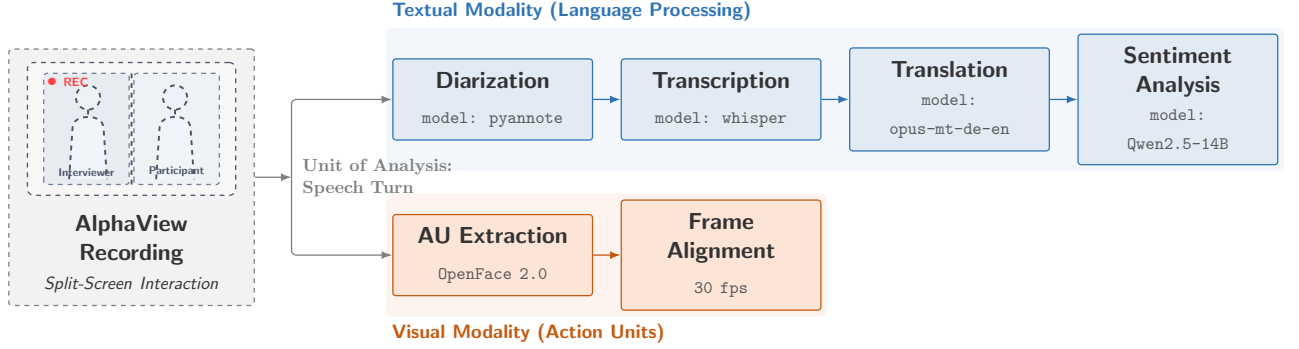


Figure 1: Data preprocessing and feature extraction pipeline

Language processing

As shown in [Figure 1](#), our language processing pipeline comprises four primary steps that transform raw audio input into sentiment-annotated speech turns (negative, neutral, positive, other).

The first step is diarization, which answers the question "who spoke when?" by outputting a Rich Transcription Time Marked (RTTM) file that denotes speaking intervals for the identified speakers. For this part, we used Pyannote 3.1 (Bredin, [2023](#); Plaquet & Bredin, [2023](#)), evaluated with the strict default parameters. Speaker diarization remains an active area of research and is known to be challenging in spontaneous conversational speech (Amorim et al., [2026](#); O’Shaughnessy, [2025](#)). Error rates on conversational speech are not negligible even for state-of-the-art systems; the maintainer-reported diarization error rate (DER) for Pyannote 3.1 is 28.5% on a telephone benchmark (Bredin & Plaquet, [2023](#); Przybocki & Martin, [2001](#)). We take this figure seriously as an indication that diarization is a genuine source of error in any pipeline built on these tools. The issue most commonly observed in our transcripts was single-speaker assignment, i.e., that the diarization could not distinguish the two speakers and assigned all intervals to one speaker (see also [Sample characteristics](#)). To validate the diarization, we reviewed the turn distribution in each transcript and removed transcripts where the diarization failed to differentiate speakers.

Second, the audio signal corresponding to the RTTM intervals is transcribed using Whisper (Radford et al., 2023). Speech turns lasting less than 1500 ms are removed as they are overwhelmingly backchannels which lack the semantic meaning we are targeting. Two heuristics determine participant or interviewer assignment: one assumes the most speaking individual is the participant, and the other assumes the first speaking individual is the interviewer. We manually checked the opening exchanges of each transcript to confirm correct assignment.

In the third step, the German transcribed speech turns were translated into English with the open-weight OPUS-MT model Helsinki-NLP/opus-mt-de-en (Tiedemann & Thottingal, 2020).

The fourth step uses an instruction-tuned open-weight LLM (Qwen2.5-14B) to assign each participant's speech turn a "positive", "negative", "neutral", or "other" label based on sentiment. We choose to work at the speech turn level as recent studies using LLMs for psychotherapy have suggested this as a fruitful unit of analysis (Bredgaard et al., 2025), and we only annotate the sentiment of the participant, not the interviewer. The LLM is provided with the preceding two speech turns as context and eight examples of correct sentiment-speech turn annotations for few-shot learning. Because the model weights are fixed and run locally, the annotation instrument does not change over time, as it would with a commercial API being continuously updated; repeated annotations of the same input are therefore close to deterministic. [Figure S1](#) in our supplementary material provides the details of our complete prompt template. The LLM is executed in the llama.cpp environment (Gerganov, 2023) for which we used the GGUF (GPT-Generated Unified Format) version of the model, specifically applying a 5-bit medium integer quantization (Q5_K_M). All steps ran fully locally on our own hardware, using a single RTX 4070 GPU. The analyses in our study have been conducted using Python 3 with Claude Opus used for coding support and code verification.

Facial movement processing

The online interviews were recorded via AlphaView in a split-screen format. To ensure accurate facial tracking, each video file was cut according to its midline, allowing the participant and interviewer to be processed independently. We then applied OpenFace 2.0 (Baltrušaitis et al.,

2016) and aligned all scores to 30 frames per second (fps), since the frame rates differed between recordings. OpenFace successfully detected and tracked facial landmarks in 99.5% of frames (mean confidence = 0.97), indicating excellent data quality throughout.

Inspired by Girard et al. (2025), our analysis focuses on smiling, operationalized through the two AUs that constitute the canonical Duchenne configuration: the cheek raiser (AU06, *orbicularis oculi*) and the lip corner puller (AU12, *zygomaticus major*) (Ekman et al., 1990). This choice leans on the probabilistic lineage of facial-expression research, in which facial movements are considered to be reliably (though far from deterministically (Barrett, 2022; Barrett et al., 2019)) associated with affective states (Cordaro et al., 2018; Keltner et al., 2019). Restricting our attention to the Duchenne smile lets us work with a well-validated, theoretically motivated signal as opposed to the full OpenFace AU set.

Following the Facial Action Coding Scheme (FACS) (Ekman & Friesen, 1978), we defined a Duchenne smile by the joint activation of AU06 and AU12, and we operationalized it in two ways. As a *presence indicator*, a frame was scored as a smile if OpenFace classified both AU06 and AU12 as present, which yields a binary signal per person and frame. As an *intensity estimate*, the smile signal was the mean of the two AU intensity scores on the 0-5 OpenFace scale, which yields a continuous signal per person and frame. These two measures are complementary. The presence indicator identifies the discrete event of a Duchenne smile, but is blind to how pronounced it is; the intensity estimate tracks the strength of the smile, including frames in which the two AUs are activated to different degrees. On the basis of this presence and intensity estimate, we use the following measures in our subsequent analyses:

Individual smiling. Individual smiling refers to each person's own signal, aggregated over the frames of the participant's speech turn: the *frequency of individual smiling* is the share of the turn's frames in which that person shows a Duchenne smile, and the *intensity of individual smiling* is the mean smile intensity over the same frames.

Frequency-based synchrony: simultaneous smiling. A frame was scored as simultaneous smiling if the presence indicator equaled one for both persons, and the *frequency of simultaneous smiling* is the share of a turn's frames scored the same way; a value of .40 thus

means that the two persons smiled at the same time during 40% of the turn.

Cross-correlation-based synchrony: cross-lagged smile synchrony. Cross-lagged smile synchrony is the dyadic version of the intensity estimate. For each participant’s speech turn, we computed the cross-lagged correlation (CLC) between the participant’s and the interviewer’s Duchenne smile intensity, evaluated for positive and negative lags of up to 2.5s in 1/30s steps. As is conventional for the CLC, the largest absolute correlation across the searched lags is taken as the synchrony of the pair (Altmann et al., 2022). This yields one coefficient per turn, which was taken as an absolute value and Fisher-z transformed. In our case, the duration of the speech turns varies considerably. We therefore computed it only for turns lasting at least four times the maximum lag, i.e. 10 s, following Cappella (1996)’s (in Schoenherr et al. (2019)) recommendation that the analyzed series be four to five times longer than the maximum lag.

Modeling

Analysis 1: Sentiment-dependent smile activation

The first core question we address is: *Do participant and interviewer smiling differ across participant speech turns labeled as negative, neutral, or positive?* The unit of analysis is the participant speech turn, where we model the distinct smile activation from each person, respectively, and for each of the two smile measures defined above; its frequency and its intensity. Because each participant contributes many turns, these observations are not independent, so we evaluate the sentiment effect with a linear mixed model (LMM) with a random intercept for participant.

Our LMM takes neutral sentiment as the reference category and specifies the smile measure as a function of sentiment,

$$\text{smile} \sim \text{sentiment} + (1 \mid \text{participant}), \quad (1)$$

and we compared its fit against a random-intercept-only model, $\text{smile} \sim 1 + (1 \mid \text{participant})$, using a likelihood-ratio test (LRT) for the omnibus effect of sentiment. The test has $df = 2$, because sentiment has three levels and the two models therefore differ by two parameters. Because

this parametric method assumes Gaussian residuals, often violated for these measures, which are bounded at zero and right-skewed, we base inference on a within-participant permutation test: sentiment labels are shuffled within each participant, both models are refitted, and the permuted LRT statistic is recorded. We rely on 3,000 such permutations per test (Phipson & Smyth, 2016).

Multiple-comparison correction uses the Benjamini-Hochberg False Discovery Rate (BH-FDR) procedure (Benjamini & Hochberg, 1995). The two operationalizations of smiling measure the same construct, so BH-FDR is applied within each operationalization across the two persons. We report the BH-FDR q -values, the LRT test statistic, and Cohen’s d (Cohen, 2013) quantifying a standardized difference between the positive and negative segments in Table 2. We conduct this analysis for both interviewer and participant smiling.

Analysis 2: Dyadic smile synchrony depending on sentiment

Second, we investigated *if dyadic smile synchrony differs according to participant sentiment*. We again take the unit of analysis as the participant’s speech turn, and we estimate synchrony with two complementary measures: the frequency of simultaneous smiling (Analysis 2a) and, as a rich literature exists on modeling synchrony using cross-correlation on AU intensity scores, the cross-lagged correlation of Duchenne smile intensity (Analysis 2b).

Analysis 2a: Frequency of simultaneous smiling. The difference between positive and negative turns was tested using a generalized linear mixed model; neutral turns are reported descriptively and were not included in the model for ease of interpretability. For each turn we counted how many of its frames showed a shared smile, out of all the frames in that turn. Our model

$$\text{shared smiling} \sim 1 + \text{sentiment} + (1 \mid \text{interviewer}) + (1 \mid \text{participant}) + (1 \mid \text{turn}), \quad (2)$$

takes negative sentiment as the reference category.

The data have two grouping levels above the speech turn: each participant contributed several turns, and several participants were interviewed by the same interviewer. Both therefore carry a random intercept. The interviewer intercept accounts for the non-independence of partic-

ipants seen by the same interviewer; the participant intercept accounts for the non-independence of turns within the same person. The third random effect, that for turn, is an observation-level random effect, i.e. one level per turn. It is required because the frames within a turn are not independent observations (a smile persists over many consecutive frames), so that without it the model would treat each of the several hundred frames of a turn as an independent Bernoulli trial and would substantially underestimate the standard error. Models were fitted with `lme4::glmer` (Bates et al., 2015), called directly from Python through `rpy2`, and compared against a null model retaining all three random effects by means of a likelihood-ratio test. We report Cohen’s d for the positive-negative difference, the LRT statistic of the sentiment effect and the associated p -value (Table 3).

Analysis 2b: Cross-lagged smile synchrony. The difference between positive and negative turns was tested using a linear mixed model. For each turn we took the largest absolute cross-lagged correlation between the participant’s and the interviewer’s smile-intensity signals and applied the Fisher- z transformation, which yields one value per turn. Our model

$$z(|r|) \sim 1 + \text{sentiment} + (1 \mid \text{interviewer}) + (1 \mid \text{participant}), \quad (3)$$

takes negative sentiment as the reference category.

As in Analysis 2a, the data have two grouping levels above the speech turn, and both carry a random intercept: the interviewer intercept accounts for the non-independence of participants seen by the same interviewer, the participant intercept for the non-independence of turns within the same person. No observation-level random effect is required here, because the cross-lagged correlation yields a single coefficient per turn rather than one observation per frame. Models were fitted with `lme4` and `lmerTest` through the Python interface `pymr4` (Jolly, 2018), and compared against a null model retaining both random effects by means of a likelihood-ratio test ($df = 1$). We report Cohen’s q , the difference between the two Fisher- z transformed correlations, the LRT statistic of the sentiment effect and the associated p -value (Table 4). Because the cross-lagged correlation requires turns of at least 10 s, Analysis 2b is based on a subset of the turns entering Analysis 2a.

Note that throughout our reporting for all analyses, the means and standard deviations in the tables are raw per-turn descriptives, computed over all turns in a sentiment category without weighting by participant. Inference rests on the mixed models described above, which account for the nesting of turns within participants and, where applicable, of participants within interviewers.

Results

Sample characteristics

Our initial sample of 83 dyads is reduced to 70, as 13 dyads could not be diarized correctly (the RTTM output assigned all speech turns to the same individual) due to high acoustic similarity. Sample characteristics are shown in [Table 1](#) and consisted predominantly of female participants (75.7%), with the majority holding a university degree (37.1%). The participants had a mean age of 27.6 years ($SD = 6.8$). The interviews were conducted by 8 female interviewers. The 13 dyads lost from the sample due to diarization failure did not differ from retained dyads in gender composition (Fisher exact $p = 0.723$).

Our sample had a total of 1598 speech turns, of which 304 were annotated as "other" and removed from analysis. The remaining 1294 turns have a roughly uniform distribution across the positive, neutral, and negative categories, as shown in [Table 1](#). [Figure 2](#) gives an overview of the core question, method, and findings achieved from the analyses conducted.

Analysis 1: Sentiment-dependent smile activation

We now report results from our first analysis examining whether the participants' and the interviewer's Duchenne smiling varied significantly according to sentiment ([Table 2](#)). Both operationalizations are reported: the frequency of smiling, i.e. the share of a turn's frames in which the person shows a Duchenne smile, and its intensity.

Participants' own smiling did not vary with the sentiment of their turns, on either operationalization (frequency: $q = .165$; intensity: $q = .414$), and the positive-negative difference was negligible in both cases. Interviewers, by contrast, smiled more than twice as often during participants' positive turns as during their negative turns (during 22.7% vs 10.3% of frames), a

Table 1: Sample characteristics: Primary sample (N = 70)

Variable	Value
Age (years), M (SD)	27.6 (6.8)
Gender, n (%)	
Female	53 (75.7%)
Male	17 (24.3%)
Education, n (%)	
Lower secondary (Hauptschule)	1 (1.4%)
Intermediate secondary (Realschule)	4 (5.7%)
Completed vocational training	6 (8.6%)
Abitur (university entrance qualification)	23 (32.9%)
Higher education degree (university)	26 (37.1%)
Missing	10 (14.3%)
In a steady relationship, n (%)	
Yes	35 (50.0%)
No	28 (40.0%)
Missing	7 (10.0%)
Speech turns, total n	1294
Speech turns by sentiment, n (%)	
Positive	444 (34.3%)
Neutral	408 (31.5%)
Negative	442 (34.2%)
Simultaneous smiling, M (SD)	0.057 (0.080)

moderate effect ($d_{\text{pos-neg}} = +.51$) that was highly significant, $\text{LR } \chi^2(2) = 29.54, q < .001$. The same pattern held for the intensity of their smiling ($M = 0.60$ in positive vs $M = 0.33$ in negative turns; $d_{\text{pos-neg}} = +.57$, $\text{LR } \chi^2(2) = 42.23, q < .001$). The two operationalizations therefore converge: sentiment was reflected in the interviewer's facial affect but not in the participant's own.

Analysis 2: Dyadic smile synchrony depending on sentiment

We now turn to our results examining if dyadic smile synchrony differs significantly across participants' negative and positive sentiment.

Analysis 2a: Frequency of simultaneous smiling. Simultaneous smiling occupied a larger share of positive than of negative turns ($M = .068, SD = .145$ vs $M = .040, SD = .118$), with neutral turns close to positive ones ($M = .065, SD = .157$). The generalized linear mixed model confirmed this difference, $\text{LR } \chi^2(1) = 23.05, p < .001$, a small effect ($d_{\text{pos-neg}} = +.21$;

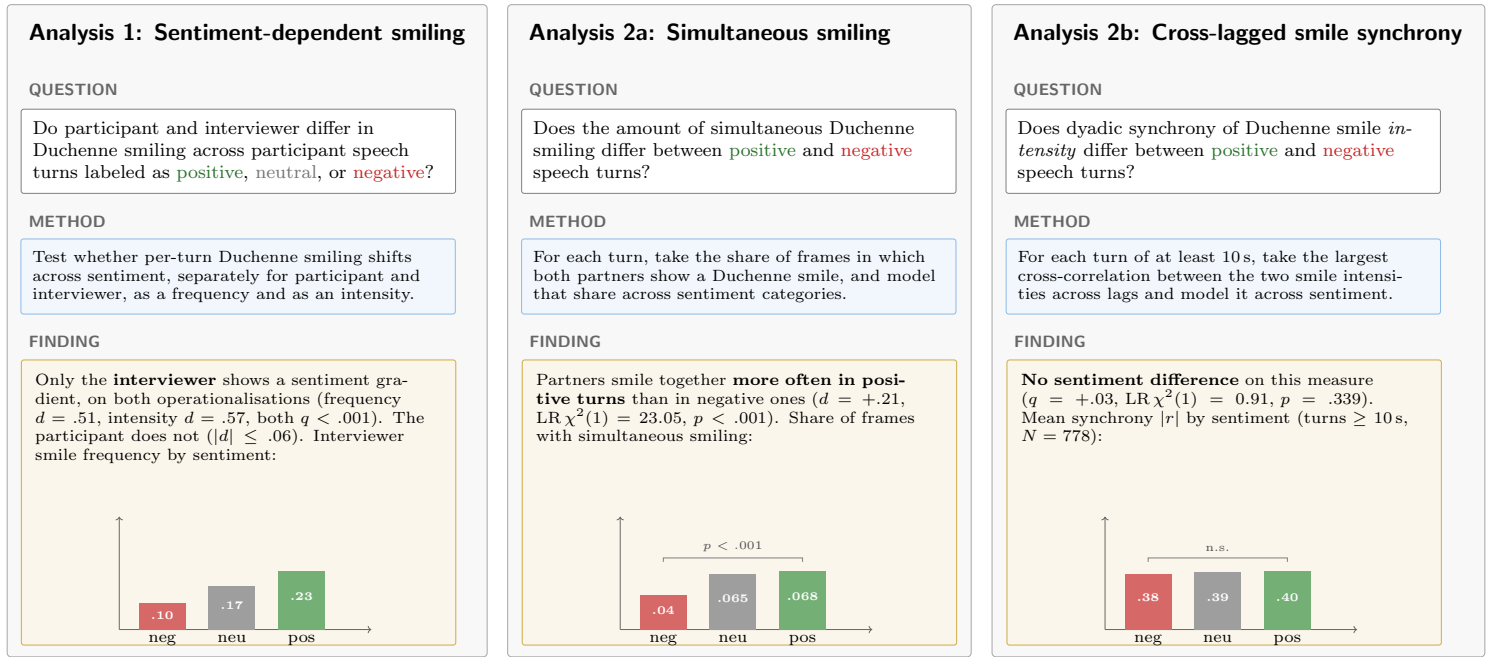


Figure 2: Roadmap of the three analyses conducted.

Table 3).

Analysis 2b: Cross-lagged smile synchrony. The CLC could be computed for 1,084 of the 1,294 turns. 210 turns were shorter than the 10 s minimum and could not be retained. As in Analysis 2a, the model was fitted on the positive and negative turns only, so that the contrast rests on 778 turns; the 306 neutral turns are reported descriptively.

Averaged across turns, the two intensity signals were weakly to moderately associated, $|r| = .40$ ($SD = .20$) in positive and $|r| = .38$ ($SD = .20$) in negative turns, with neutral turns

Table 2: Individual Duchenne smiling by participant speech-turn sentiment, for the participant and the interviewer, by two operationalisations: frequency and intensity.

Person	Measure	Positive <i>M</i> (SD)	Neutral <i>M</i> (SD)	Negative <i>M</i> (SD)	$d_{\text{pos-neg}}$	$\text{LR } \chi^2(2)$	q
Participant	Frequency	0.132 (0.201)	0.144 (0.234)	0.122 (0.192)	+.05	3.31	.165
Interviewer	Frequency	0.227 (0.280)	0.170 (0.268)	0.103 (0.202)	+.51	29.54	<.001
Participant	Intensity	0.393 (0.359)	0.412 (0.419)	0.412 (0.355)	-.06	1.76	.414
Interviewer	Intensity	0.603 (0.550)	0.451 (0.530)	0.332 (0.389)	+.57	42.23	<.001

Note. Frequency refers to the share of a turn's frames showing an individual Duchenne smile, *intensity* to the mean Duchenne smile activation. *M* (SD) are raw per-turn descriptives, computed over all turns in a sentiment category without weighting; the test statistics beside them come from the mixed models, which account for the nesting of turns within participants. Cohen's *d* is the positive–negative difference (positive *d* = more smiling in positive turns). *q* = BH-FDR corrected *p*-value.

Table 3: Analysis 2a: Frequency of simultaneous Duchenne smiling by participant speech-turn sentiment.

<i>N</i>	<i>M</i> (SD)			$d_{\text{pos-neg}}$	LR $\chi^2(1)$	<i>p</i>
	Positive	Neutral	Negative			
886	0.068 (0.145)	0.065 (0.157)	0.040 (0.118)	+.21	23.05	<.001

Note. *M* (SD) are per-turn frequencies of simultaneous Duchenne smiling, i.e. the share of frames in which participant and interviewer both smile. *N* = positive and negative turns entered into the model; neutral turns are descriptive.

in between ($|r| = .39$, $SD = .20$; Table 4). The difference between positive and negative turns is very small and the linear mixed model did not confirm it, LR $\chi^2(1) = 0.91$, $p = .339$, $q_{\text{pos-neg}} = +.03$. The cross-lagged correlation between the two partners' smile-intensity signals thus showed no such difference.

Table 4: Analysis 2b. Synchrony of Duchenne-smile intensity by participant speech-turn sentiment.

<i>N</i>	Positive	Neutral	Negative	$q_{\text{pos-neg}}$	LR $\chi^2(1)$	<i>p</i>
	<i>M</i> (SD)	<i>M</i> (SD)	<i>M</i> (SD)			
778	.405 (.196)	.386 (.204)	.378 (.198)	+.03	0.91	.339

Note. *M* (SD) are the per-turn synchrony correlations $|r|$, computed over all turns in a sentiment category. *N* = positive and negative turns of at least 10 s entered into the model. Linear mixed model on the Fisher-*z* transformed coefficients, with random intercepts for interviewer and participant; negative sentiment = reference. Cohen's *q* is the positive–negative difference between the two Fisher-*z* transformed correlations, the appropriate effect size for comparing correlations; it is therefore not the difference between the two means in the table.

Discussion

In this study, we presented a new approach to contextualize nonverbal synchrony, whose benefits we illustrated through analyses of an attachment interview. Across the speech turns retained for our analyses, annotated by an open-weight LLM running entirely on local hardware, two main findings emerged. First, only the interviewer's Duchenne smiling varied with the sentiment of the participant's speech, whereas the participants' own smiling did not. Second, the two partners smiled simultaneously more often during positive than during negative turns.

Interpreting the interviewer's smiling behavior

Girard et al. (2021) have argued that the meaning of the Duchenne configuration depends on the situation in which it occurs and on how it is perceived. We find that interviewers smiled more often and more intensely while participants spoke positively (Table 2), and both partners smiled at the same time more often during these turns (Table 3). This synchrony is cross-modal: the participant expresses the sentiment verbally, and the interviewer responds with a smile. It seems that the interviewer mirrors the sentiment of the participant, possibly to signal understanding and to encourage the participant to continue speaking. The reduced smiling during negative turns can be interpreted in the same way, since a smile that persists while a participant recalls a distressing experience would likely be perceived as misattunement rather than warmth. Our findings of the interviewer's behavior can be interpreted in the light of the reading by Csibra and Gergely (2009). Here, it is described how direct gaze, eyebrow raise, and smiling function as signals that mark a behavior as addressed to a particular recipient. According to this view, the interviewer's smiling functions as an ostensive cue, signaling to the participant that he/she is being attended to.

Comparison to existing findings in the sentiment-related literature

The direction of our effect agrees with the studies highlighted in our introduction (Altmann, 2011; Kimura & Daibo, 2006; Kykyri et al., 2025; Tschacher et al., 2014): synchrony was more pronounced when the talk was positive. Tschacher et al. (2014) report that their fun

task, the condition with the lowest negative affect, showed higher synchrony than competition ($d = 0.30$) and cooperation ($d = 0.49$), against $d = +.21$ for our positive–negative difference in simultaneous smiling. We thus recover an effect of the same order as the established literature, but with a design considerably better suited to the question in two primary respects.

The first is the modality. Previous studies all quantified movement synchrony, whether from motion energy (Altmann, 2011; Kykyri et al., 2025; Tschacher et al., 2014) or from observer ratings of coordinated movement (Kimura & Daibo, 2006). Body movement is a broad channel of display whose relation to sentiment is indirect. The facial channel we analyze is more directly linked to affective display than body movement, and is therefore the more appropriate measure for a question about sentiment.

The second difference relates to how sentiment was operationalized. Kimura and Daibo (2006) assigned dyads to talk for three minutes about the happiest or the saddest events of their recent days; Tschacher et al. (2014) prescribed cooperative, competitive and fun-task instructions for interactions of five minutes; Altmann (2011) created neutral and conflict situations with a computer game; and Kykyri et al. (2025) had the sentiment of episodes coded from the recording by a researcher. In each case, sentiment remained fixed for a whole episode, whether by experimental instruction or by observer coding. Synchrony can then be compared between episodes, but not within one, and the pulls that Gordon et al. (2025) describe remain unobserved. Our unit is the participant’s speech turn, whose duration is markedly more fine-grained than full episodes, and its sentiment arises from organic developments in the conversation itself. Our approach is therefore more ecologically valid to psychotherapy research, as it permits the study of sentiment-specific synchrony without imposing an artificial constraint on participants.

This is where we believe the present approach contributes to the theory. Sentiment annotation of speech turns characterizes context from within the interaction itself, at the resolution at which the pulls are assumed to change. Two consequences follow. First, flexibility can be defined with respect to moment-to-moment changes in valence. Sayda et al. (2026) operationalized it as the degree to which synchrony fluctuated over the course of a session, which registers that synchrony changed but not what it changed in response to. Conditioning synchrony on the sentiment of each turn adds that. Second, since the annotation is produced

automatically, the characterization is reproducible across studies and can be applied to any recorded conversation without additional coding effort. We therefore regard turn-level sentiment annotation as a practicable operationalization of the context component of the theory (Gordon et al., 2025).

The applicability of language modeling for sentiment classification in psychotherapy research

Our contribution rests on the core assumption that the sentiment annotations provided by our chosen, open-weight LLM (Qwen-14b, Figure 1) have been performed at a level that renders the analyses and clinical insights of our study valid and relevant. Multiple arguments exist to support this assumption.

First, the broader feasibility and construct validity of automated sentiment analysis in psychotherapy have already been established, demonstrating strong alignment between the model and human clinical judgments (Eberhardt, Schaffrath, et al., 2025). Building on this foundation, for a relatively simple annotation task such as sentiment classification, studies suggest that LLMs in the small-to-mid parameter range are satisfactory (Nasution et al., 2025) and can be improved substantially via prompt engineering (Petersen et al., 2025), such as the few-shot learning implemented in our study (Figure S1). Further, by relying on just three sentiment categories, we reduce annotation complexity to a level well within the documented capability of models in the Qwen2.5-14B class.

Second, our annotation pipeline was developed iteratively. Throughout the architecture design process, we would continuously inspect model outputs and adjust the prompt template and few-shot examples until the labels were consistent with our clinical understanding. While this evaluation was not formalized as an inter-rater reliability analysis, a limitation we acknowledge, it was not a process in which the model's outputs went uninspected.

Third, as our study contains highly sensitive data, sending this to cloud-based APIs (such as OpenAI or Anthropic) for processing with state-of-the-art models, would introduce significant risks regarding GDPR compliance, data-use agreements, and patient confidentiality (Abdou et al., 2025; Eberhardt, Vehlen, et al., 2025). Having this processing take place on strictly

local hardware is far more secure and privacy-aware, and is becoming increasingly feasible using quantization and efficient environments (Gerganov, 2023). No data leave the machine on which they are already stored, so the annotation raises none of the confidentiality questions that accompany ChatGPT or comparable cloud-based services. Our pipeline thus shows how psychotherapy researchers working in low-compute environments can benefit from the powerful capabilities of modern LLMs.

Finally, we believe that advances in AI-assisted coding are democratizing the skill set required to run the pipeline presented here, thus enabling our approach for the broader psychotherapy research community. The primary limiting factor is not writing code, but specifying its requirements and verifying it. These are largely methodological judgments, appropriate for the expertise gathered at a psychology degree.

Flexibility of the approach

The approach presented here is not tied to the attachment interview or to our choice of sentiment categories. It requires only a recording of a conversation between two people, so the same pipeline applies to therapy sessions, intake interviews, or any other dyadic clinical conversation. Researchers have complete freedom to change the annotation scheme between settings: the three sentiment categories used here are specified in the prompt (Figure S1) and can be replaced with whichever categories the research question calls for, for instance categories drawn from a particular theory of the therapeutic process. The facial analysis is unaffected by that choice, since sentiment enters it only as the label of the speech turn. Together with the local processing described above, this makes the approach applicable to the kinds of sensitive recordings that psychotherapy research typically involves.

Our own design choice, to prompt the LLM to annotate speech turns as positive, neutral, or negative, is ultimately the result of a delicate balancing act between content expressivity and analytical feasibility, in which we have prioritized the latter. Highly rewarding research designs could arguably be formulated by prioritizing expressivity, for instance, by analyzing contingent facial movements within the dyad at the level of each word. A higher degree of expressivity could also be achieved at the level of the speech turn, by prompting the LLM to choose among

a battery of theories to annotate the specific turn. Such advances, however, are fundamentally limited by the size of the data corpora commonly available in the psychotherapy research communities. In light of these constraints, delimiting speech turns into a triadic sentiment bucket provides a gross yet informative and tractable framework for understanding the meaning of nonverbal processes during treatment.

Limitations and future research

We would like to note a series of limitations to our study. First, the generalizability of the findings is limited. The sample was small and mostly non-clinical, so the extent to which the observed sentiment-dependent facial synchrony patterns transfer to clinical dyads and to sustained treatment relationships remains an open question.

The second level concerns the synchrony calculation itself. There is still poor consensus on which algorithm is best suited to a given research question: cross-lagged correlation, cross-lagged regression, recurrence-based and entropy-based measures capture partly distinct facets of synchrony and show limited convergent validity, such that the same data can yield substantively different and occasionally contradictory conclusions depending on the measure chosen (Altmann et al., 2022; Schoenherr et al., 2019). The present study is not exempt: across the two measures applied here (Analyses 2a and 2b, Table 3 and Table 4) only the frequency measure yielded a significant sentiment effect, though both pointed in the same direction. Our findings thus echo those reported in other studies comparing frequency to cross-correlation based synchrony measures (Schoenherr et al., 2019).

Ultimately, our contribution demonstrates that leveraging LLMs to contextualize the spoken content reveals context-dependent nonverbal processes, offering a scalable first step toward the context-sensitive, flexibility-oriented measurement that both theory and methodology increasingly call for (Barrett, 2022; Gordon et al., 2025).

Declarations

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Conflict of Interest

The authors report no conflicts of interest.

Ethics Approval

The ethics committee of the Medical School Berlin, Berlin, Germany, approved the study (file number MSB-2023/122).

Availability of data and materials

The data analyzed in the current study cannot be made publicly available due to the sensitive nature of the interview recordings and the terms of the informed consent provided by participants.

Code availability

The code for our preprocessing pipeline and analyses is publicly available at github.com/mltrinhammer/llm_smiling.

Supplementary Material

Our supplementary material is available online at our github repository, together with the preprocessing pipeline and analyses.

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